PROBLEM 42

Binet + Effective Potential

# Particle Attracted to the Origin by $F = m\mu/r^{\alpha}$

Original statement · Problems.pdf, p. 29

*(Advanced problem.)* A point particle $P$ of mass $m$ is attracted by the origin of coordinates $O$ of the inertial reference frame $S_0$ by a force with magnitude

$$F = \frac{m\mu}{r^{\alpha}}$$

where $r$ is the distance between $O$ and $P$, and $\mu$, $\alpha$ are known positive constants.

**Notation.** The official text writes $F = m\alpha/r^{\beta}$ (strength $\alpha$, exponent $\beta$) and asks for $\beta = 1, 2, 3, 4$. Here the strength is called $\mu$ and the exponent $\alpha$, so the cases "$\alpha = 1, 2, 3, 4$" below are the official $\beta = 1, 2, 3, 4$, and $\mu/h^2$ is the official $\alpha/h^2$.

**(a)** Show that the angular momentum vector and the mechanical energy of $P$ are conserved quantities of motion, and that the motion of $P$ occurs in a plane.

**(b)** Using polar coordinates $(r, \theta)$ in the plane of motion, find a second-order differential equation to solve for $u \equiv 1/r$ as a function of $\theta$ (i.e., $u = u(\theta)$).

**(c)** Find a first integral of this equation and discuss the limits of motion of $u$ as a function of $\mu/h^2$, $\alpha$, and the values of $u$ and $u' \equiv du/d\theta$ at $t = 0$, where $h$ is the magnitude of the angular momentum vector about the origin per unit mass. For $\alpha = 1, 2, 3$, and $4$, determine under which conditions: (i) the particle escapes to infinity ($u \to 0$); (ii) the particle reaches $r = 0$ ($u \to \infty$); (iii) the particle oscillates between two values of $u$.

*Hint: try to solve first the problem with $\alpha = 2$, and then try the other values.*

**Worked Solution (4-Phase Methodology)**

PHASE 1

## Physical Statement, Hypotheses & Parameters

- **System:** a particle $P$ of mass $m$ attracted to $O$ by $\mathbf{F} = -\dfrac{m\mu}{r^{\alpha}}\mathbf{e}_r$.
- **Hypotheses:** $S_0$ inertial; $\mu > 0, \alpha > 0$ constants; no other forces; $h \neq 0$ unless stated.
- **Degrees of freedom:** free particle, $\mathrm{CDOF} = 3$; after part (a) the motion is planar, $\mathrm{CDOF} = 2$ with coordinates $(r, \theta)$.
- **Variables:** $u(\theta) = 1/r(\theta), u' = du/d\theta, h = \|\mathbf{r} \times \mathbf{v}\|$ [m$^2$/s].

| Symbol | Meaning | SI unit |
|---|---|---|
| $\mu$ | force strength per unit mass, $F/m = \mu/r^{\alpha}$ | $\mathrm{m}^{\alpha+1}/\mathrm{s}^2$ |
| $h$ | specific angular momentum | $\mathrm{m}^2/\mathrm{s}$ |
| $u$ | reciprocal radius | $\mathrm{m}^{-1}$ |
| $\xi$ | specific mechanical energy | J/kg = $\mathrm{m}^2/\mathrm{s}^2$ |

PHASE 2

## Reference Frames, Vector Bases & Coordinate Geometry

- **Change of basis (polar basis in the plane of motion).** Take $\mathbf{k}$ along $\mathbf{H}_O$. The polar basis $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis $\mathcal{B}_0 = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ rotated by $\theta(t)$ about $\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\theta + \sin^2\theta = 1, \qquad [{}_0R_1]^T[{}_0R_1] = I$$

The angular velocity of $\mathcal{B}_1$ is $\boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{k}$; Poisson's formula $\dot{\mathbf{e}} = \boldsymbol{\omega} \times \mathbf{e}$ gives $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_r$. The product rule on $\mathbf{r} = r\mathbf{e}_r$ yields

$$\mathbf{v} = \dot{r}\,\mathbf{e}_r + r\dot{\theta}\,\mathbf{e}_\theta, \qquad \mathbf{a} = \left(\ddot{r} - r\dot{\theta}^2\right)\mathbf{e}_r + \left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right)\mathbf{e}_\theta$$

PHASE 3

## Step-by-Step Mathematical Deduction

### Part (a): conservation laws and planar motion

**Pedagogical Justification:** The force line of action passes through the fixed point $O$, so its torque about $O$ vanishes, and the angular momentum theorem about a fixed point (Notes Eq. 4.53) gives a constant $\mathbf{H}_O$.

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = r\mathbf{e}_r \times \left(-\frac{m\mu}{r^\alpha}\mathbf{e}_r\right) = \mathbf{0} \implies \left.\frac{d\mathbf{H}_O}{dt}\right|_0 = \mathbf{0} \implies \mathbf{H}_O = m\,\mathbf{r} \times \mathbf{v} = \text{const}$$

Since $\mathbf{r} \cdot \mathbf{H}_O \equiv 0$ and $\mathbf{v} \cdot \mathbf{H}_O \equiv 0$, $P$ stays in the plane through $O$ perpendicular to $\mathbf{H}_O$.

**Pedagogical Justification:** The force is purely radial and depends only on $r$, so its elementary work $\mathbf{F} \cdot d\mathbf{r} = F(r)\,dr$ is the exact differential of a function of $r$ (equivalently $\nabla \times \mathbf{F} = \mathbf{0}$). The force is conservative, so the energy is conserved. The potential energy per unit mass follows from Barrow's rule with the reference at infinity (for $\alpha > 1$; for $\alpha \le 1$ the reference point is arbitrary):

$$V_0(r) = -\int_\infty^r \left(-\frac{\mu}{s^\alpha}\right) ds = \mu\left[\frac{s^{1-\alpha}}{1-\alpha}\right]_\infty^r = -\frac{\mu}{(\alpha-1)\,r^{\alpha-1}} \quad (\alpha > 1), \qquad V_0(r) = \mu \ln r \quad (\alpha = 1)$$

$$\xi = \tfrac{1}{2}v^2 + V_0(r) = \text{const}$$

**Part (b): Binet equation for a power law**

**Pedagogical Justification:** Eliminate time with $h = r^2\dot{\theta} = \text{const}$, i.e. $\dot{\theta} = hu^2$. The chain rule for $r = u^{-1}$ gives $dr/d\theta = -u'/u^2$.

$$\dot{r} = \frac{dr}{d\theta}\dot{\theta} = \left(-\frac{u'}{u^2}\right)\left(hu^2\right) = -hu', \qquad \ddot{r} = \frac{d}{dt}\left(-hu'\right) = -hu''\,\dot{\theta} = -h^2u^2u''$$

The radial equation $\ddot{r} - r\dot{\theta}^2 = -\mu/r^\alpha = -\mu u^\alpha$ with $r\dot{\theta}^2 = \frac{1}{u}(hu^2)^2 = h^2u^3$ becomes

$$-h^2u^2u'' - h^2u^3 = -\mu u^\alpha \quad \overset{\div\,(-h^2u^2)}{\Longrightarrow} \quad \boxed{u'' + u = \frac{\mu}{h^2}\,u^{\alpha-2}}$$

**Part (c): first integral**

**Pedagogical Justification:** Write the specific energy with $\dot{r} = -hu'$ and $r\dot{\theta} = hu$. Since $\xi$ is constant, this is a first integral; it can be checked by multiplying the Binet equation by $u'$ and integrating in $\theta$.

$$\xi = \tfrac{1}{2}h^2\left[(u')^2 + u^2\right] + V_0\!\left(\frac{1}{u}\right) \implies \boxed{(u')^2 + W_{\text{eff}}(u) = E^*}, \quad E^* \equiv \frac{2\xi}{h^2}, \quad W_{\text{eff}}(u) = u^2 + \frac{2V_0(1/u)}{h^2}$$

Check: $\frac{d}{d\theta}\left[(u')^2 + W_{\text{eff}}\right] = 2u'u'' + W'_{\text{eff}}u' = 0$ gives $u'' = -\frac{1}{2}W'_{\text{eff}}$, which is the Binet equation. Motion exists only where $W_{\text{eff}}(u) \le E^*$; turning points ($u' = 0$) are the roots of $W_{\text{eff}}(u) = E^*$, and a minimum or a maximum of $W_{\text{eff}}$ is a circular orbit.

**Case $\alpha = 2$: inverse square (Kepler)**

- Binet: $u'' + u = \mu/h^2$. Potential: $V_0 = -\mu u$, so $W_{\text{eff}} = u^2 - \dfrac{2\mu}{h^2}u = \left(u - \dfrac{\mu}{h^2}\right)^2 - \left(\dfrac{\mu}{h^2}\right)^2$, a parabola with minimum $-(\mu/h^2)^2$ at $u = \mu/h^2$.
- **(i) Escape ($u \to 0$):** possible if $E^* \ge W_{\text{eff}}(0) = 0$, i.e. $\xi \ge 0$ (parabola or hyperbola).
- **(ii) Reaching $r = 0$:** impossible for $h \neq 0$, because $W_{\text{eff}} \to +\infty$ as $u \to \infty$.
- **(iii) Oscillation:** for $-(\mu/h^2)^2 < E^* < 0$, i.e. $-\mu^2/(2h^2) < \xi < 0$: $u$ oscillates between the two roots (periapsis and apoapsis of an ellipse); $E^* = -(\mu/h^2)^2$ is the circular orbit.

**Case $\alpha = 1$: $F \propto 1/r$**

- Binet: $u'' + u = \dfrac{\mu}{h^2 u}$. Potential $V_0(1/u) = -\mu \ln u$, so $W_{\text{eff}} = u^2 - \dfrac{2\mu}{h^2} \ln u$.
- $W_{\text{eff}} \to +\infty$ both as $u \to 0$ (because $-\ln u \to +\infty$) and as $u \to \infty$. Its minimum is at $2u - \dfrac{2\mu}{h^2 u} = 0$, i.e. $u_0 = \sqrt{\mu}/h$.
- **(i)** Escape is impossible for any finite energy. **(ii)** Reaching $r = 0$ is impossible. **(iii)** Every orbit with $E^* > W_{\text{eff}}(u_0)$ oscillates between two finite turning points; $E^* = W_{\text{eff}}(u_0)$ is the circular orbit $r = h/\sqrt{\mu}$.

**Case $\alpha = 3$: $F \propto 1/r^3$**

- $V_0 = -\dfrac{\mu}{2r^2} = -\dfrac{\mu}{2} u^2$. Binet: $u'' + \lambda u = 0$ with $\lambda \equiv 1 - \mu/h^2$. Then $W_{\text{eff}} = \lambda u^2$ and $(u')^2 + \lambda u^2 = E^*$.
- **$h^2 > \mu$ ($\lambda = \kappa^2 > 0$, $\kappa = \sqrt{1 - \mu/h^2}$):** $u = A\cos(\kappa\theta + \delta)$ with $A = \sqrt{u_0^2 + (u_0'/\kappa)^2}$. The cosine vanishes at a finite angle, so $u \to 0$ in every case: **(i) the particle always escapes**; a particle moving inward ($u_0' > 0$) is first reflected at the periapsis $u = A$. **(ii)** never, since $u \le A$. **(iii)** no oscillation between two positive values.
- **$h^2 < \mu$ ($\lambda = -\kappa^2 < 0$, $\kappa = \sqrt{\mu/h^2 - 1}$):** $u = C_1 e^{\kappa\theta} + C_2 e^{-\kappa\theta}$ with $C_1 = \frac{1}{2}(u_0 + u_0'/\kappa)$, $C_2 = \frac{1}{2}(u_0 - u_0'/\kappa)$. If $C_1 > 0$ (i.e. $u_0' > -\kappa u_0$), $u \to \infty$: **(ii) the particle reaches the origin**. If $C_1 < 0$ ($u_0' < -\kappa u_0$), $u$ reaches 0 at a finite angle: **(i) escape**. If $C_1 = 0$, $u = u_0 e^{-\kappa\theta} \to 0$ only asymptotically (spiral $r = r_0 e^{\kappa\theta}$), a limiting case of escape. **(iii)** no oscillation.
- **$h^2 = \mu$ ($\lambda = 0$):** $u'' = 0 \Rightarrow u = u_0 + u_0'\theta$ (reciprocal spiral). $u_0' > 0$: reaches the origin; $u_0' < 0$: escapes at $\theta = u_0/|u_0'|$; $u_0' = 0$: circle $r = 1/u_0$.

**Case $\alpha = 4$: $F \propto 1/r^4$**

- Binet: $u'' + u = \dfrac{\mu}{h^2} u^2$. $V_0 = -\dfrac{\mu}{3} u^3$, so $W_{\text{eff}} = u^2 - \dfrac{2\mu}{3h^2} u^3 = u^2\left(1 - \dfrac{2\mu}{3h^2} u\right)$.
- $W_{\text{eff}}'(u) = 2u - \dfrac{2\mu}{h^2} u^2 = 2u\left(1 - \dfrac{\mu}{h^2} u\right)$: stationary points $u = 0$ (minimum, $W = 0$) and $u_b = h^2/\mu$ (maximum, barrier $W_{\max} = u_b^2 - \dfrac{2\mu}{3h^2} u_b^3 = \dfrac{h^4}{3\mu^2}$). For $u > u_b$, $W_{\text{eff}}$ decreases without bound and crosses zero at $u_0 = 3h^2/(2\mu) > u_b$.
- **There is no interior minimum**, so the allowed set $\{W_{\text{eff}} \le E^*\}$ either contains $u = 0$ or extends to $u \to \infty$. Because $\dot\theta = h u^2 > 0$, $u'$ has the sign of $\dot u$.
- **$0 \le E^* < W_{\max}$:** two roots $u_1 \in [0, u_b)$ and $u_2 > u_b$; allowed $u \le u_1$ or $u \ge u_2$. If $u(0) \le u_1$ the particle is reflected at $u_1$ and escapes; if $u(0) \ge u_2$ it is reflected at $u_2$ and falls to the origin.
- **$E^* < 0$:** only $u \ge u_2 > u_0$ is allowed; the particle reaches $r = 0$. **$E^* > W_{\max}$:** no turning points; $u'(0) < 0$ (moving outward) escapes and $u'(0) > 0$ (moving inward) falls to the origin. **$E^* = W_{\max}$:** the unstable circular orbit $u = u_b$ and asymptotic approaches to it.

- **(i)** escape: $0 \le E^* < W_{\max}$ with $u(0) < u_b$, or $E^* > W_{\max}$ with $u'(0) < 0$. **(ii)** reaching $r = 0$: $E^* < 0$, or $0 \le E^* < W_{\max}$ with $u(0) > u_b$, or $E^* > W_{\max}$ with $u'(0) > 0$. **(iii)** bounded oscillation between two values of $u$: **does not occur**; it would need an interior minimum of $W_{\text{eff}}$, and the only bounded motion is the unstable circle $u = u_b$.

PHASE 4

### Physical Interpretation & Dimensional Verification

- **Dimensional check:** $[F/m] = \mathrm{m/s^2} = [\mu]/\mathrm{m}^\alpha$ gives $[\mu] = \mathrm{m}^{\alpha+1}/\mathrm{s}^2$. With $[h^2] = \mathrm{m}^4/\mathrm{s}^2$ and $[u] = \mathrm{m}^{-1}$, $\dfrac{\mu}{h^2}u^{\alpha-2}$ has units $\mathrm{m}^{\alpha-3}\,\mathrm{m}^{-(\alpha-2)} = \mathrm{m}^{-1}$, the same as $u$ and $u''$ ($\theta$ is dimensionless).
- **Stability of circular orbits (consequence of the first integral):** for general $\alpha$, $W'_{\text{eff}}(u_0) = 0$ gives $u_0^{\alpha-3} = h^2/\mu$ and $W''_{\text{eff}}(u_0) = 2(3-\alpha)$, so the circular orbit is a minimum (stable) for $\alpha < 3$ and a maximum (unstable) for $\alpha > 3$, in agreement with the cases above ($\alpha = 4$: barrier at $u_b = h^2/\mu$).
- **Numerical check** (Python, RK4 with step $10^{-4}$ rad on $u'' = -u + (\mu/h^2)u^{\alpha-2}$, units with $\mu = h = 1$ unless stated):

| Case | Initial data $(u_0, u_0')$ | Predicted | Integration result |
|---|---|---|---|
| $\alpha = 2$ | $(1.5, 0)$ | bounded, 0.5 to 1.5 | $u \in [0.5000, 1.5000]$ over 60 rad |
| $\alpha = 2$ | $(0.5, -0.9)$, $E^* > 0$ | escape | $u = 0$ at $\theta = 0.823$ |
| $\alpha = 4$ | $(0.5, 0)$, $E^* = 1/6 < 1/3$ | reflected, then escape | $u = 0$ at $\theta = 2.078$ |
| $\alpha = 4$ | $(1.2, 0)$, $E^* = 0.288 < 1/3$ | collapse | $u \to \infty$ at $\theta = 3.713$ |
| $\alpha = 4$ | $(0.5, 0.9)$, $E^* > 1/3$ | collapse | $u \to \infty$ at $\theta = 3.258$ |
| $\alpha = 4$ | $(0.5, -0.9)$, $E^* > 1/3$ | escape | $u = 0$ at $\theta = 0.524$ |
| $\alpha = 4$ | $(1, 0)$, $E^* = W_{\max}$ | unstable circular orbit | $u = 1$ for 20 rad |
| $\alpha = 3$, $\mu/h^2 = 2$ | $(1, 0.2)$, $C_1 > 0$ | collapse | $u = \cosh\theta + 0.2\sinh\theta$ is finite at every finite $\theta$ and $\to \infty$ only as $\theta \to \infty$ (the particle reaches $r = 0$ in finite time after infinitely many turns); $\theta = 4.423$ is the integrator cutoff |

| Case | Initial data $(u_0, u_0')$ | Predicted | Integration result |
|---|---|---|---|
| $\alpha = 3$, $\mu/h^2 = 2$ | $(1, -1.5)$, $C_1 < 0$ | escape | $u = 0$ at $\theta = 0.805$ |
| $\alpha = 3$, $\mu/h^2 = 1/2$ | $(1, 0)$ | escape | $u = 0$ at $\theta = 2.222$ |
| $\alpha = 3$, $\mu = h^2$ | $(1, -0.1)$ | escape at $\theta = 10$ | $u = 0$ at $\theta = 10.000$ |
| $\alpha = 1$ | $(0.5, 0.3)$ | bounded | $u \in [0.471, 1.653]$ over 100 rad |

The official key (Problems.pdf) gives no answer for Problem 42, so these checks are against the derived classification only.

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*Source: Problems.pdf (PDF page 29); no key available*